

PhenoLeaf-TS

A Time-Series Benchmark for Leaf Instance Segmentation, Tracking, and Growth Stage Classification

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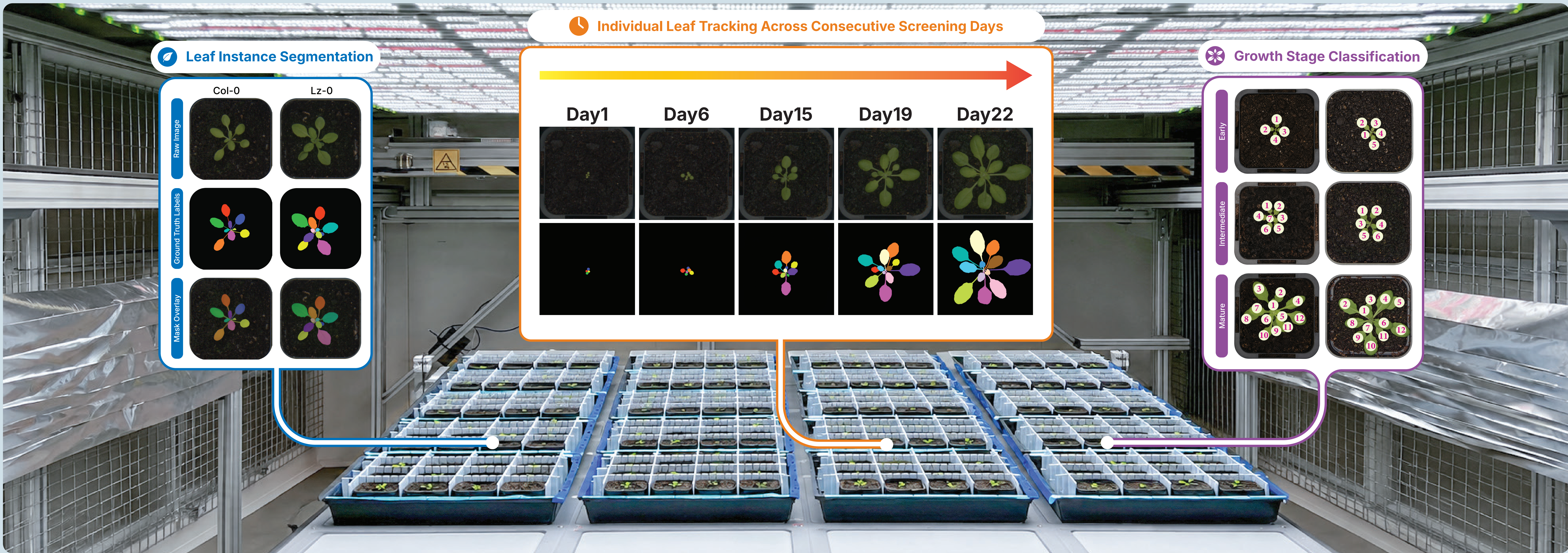


Problem & Motivation

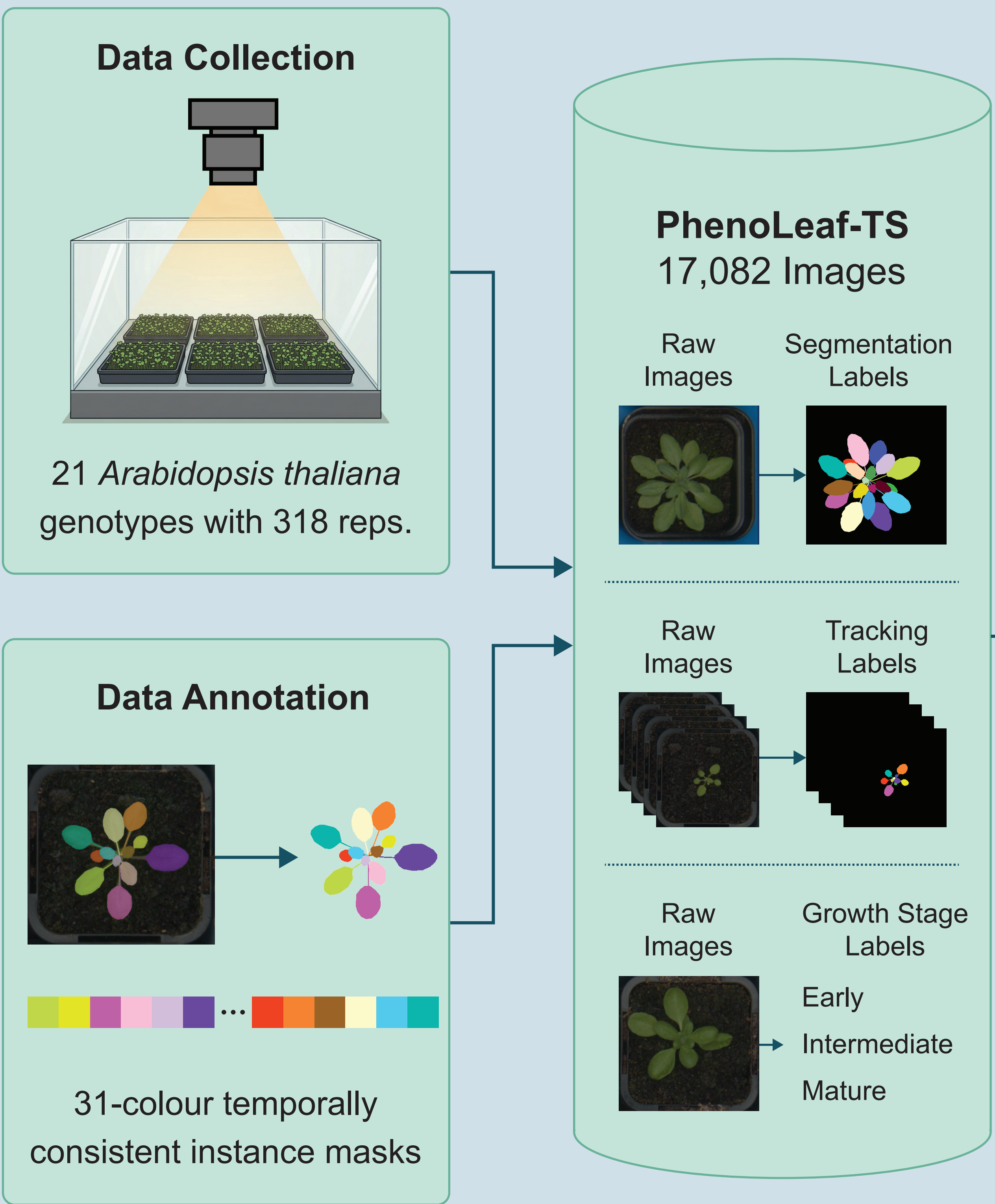
- Quantifying leaf-level traits is essential for image-based high-throughput plant phenotyping (HTPP), but requires correct **leaf segmentation** and **precise tracking** throughout growth despite overlap, occlusion, and morphology changes.
- Existing datasets are mostly **small, single-time-point collections** or **lack persistent leaf identities**, preventing joint evaluation of segmentation, tracking, and growth-stage classification.

Existing Datasets	PhenoLeaf-TS
⚠️ Limited scale and diversity	✅ 17,082 images; 21 genotypes
⚠️ Limited temporal coverage	✅ 318 plant image sequences
⚠️ Limited leaf identity annotations	✅ Consistent leaf identities
⚠️ Limited joint evaluation	✅ Three benchmark tasks

💡 A single dataset for evaluating leaf segmentation, tracking, and growth stage classification together.



STAGE 1: Dataset Preparation



STAGE 2: Benchmark Computer Vision Tasks

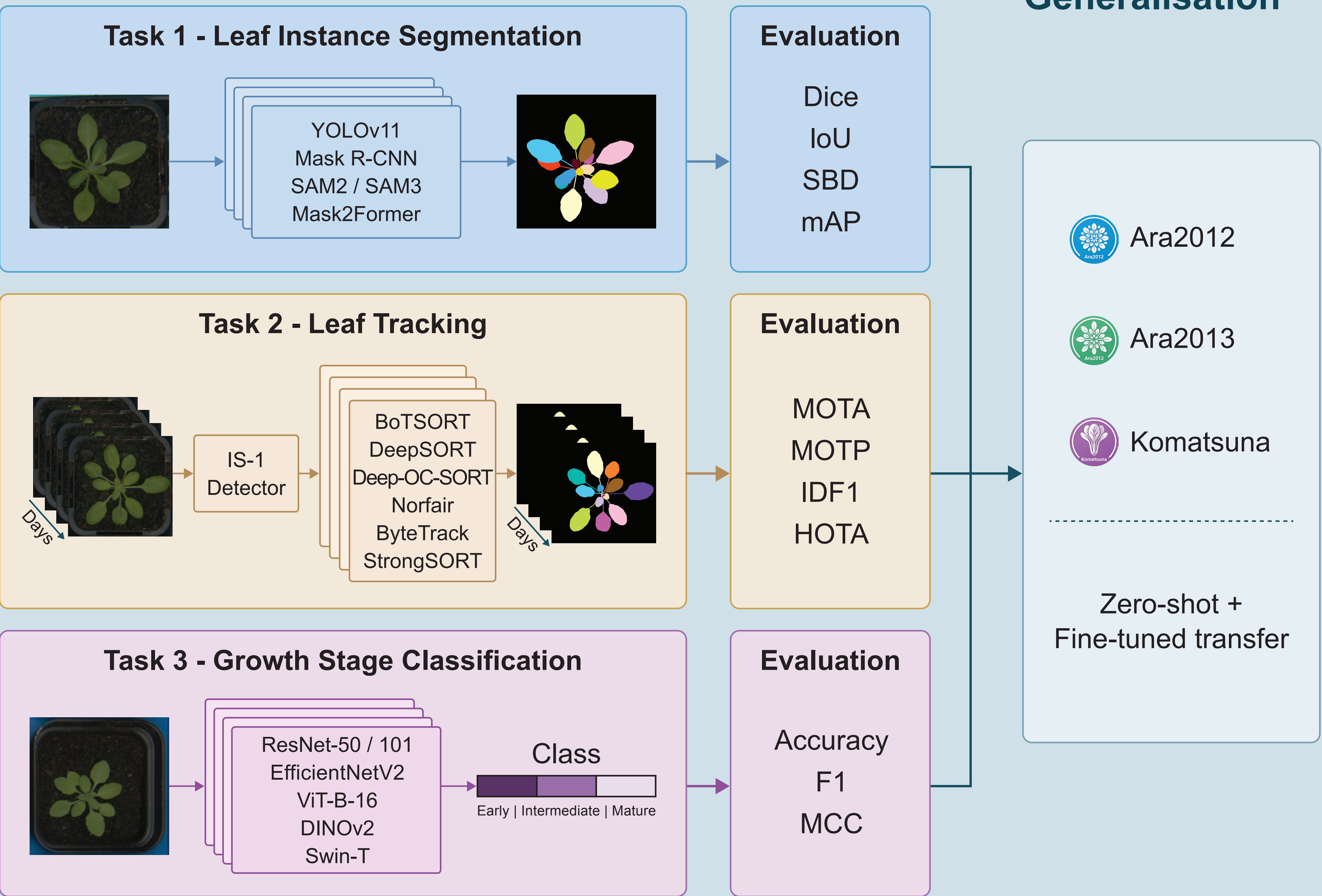


Table 1: Selected benchmark results for leaf instance segmentation, tracking and growth stage classification on PhenoLeaf-TS.

ID	Model	Backbone	mAP	mAP ₅₀	mAP ₇₅
IS-1	YOLOv11-seg	CSPDarknet-X	68.2	84.7	--
IS-3	Mask R-CNN	R50-FPN	73.2	94.3	82.9
IS-8	Mask2Former	Swin-L	71.9	94.6	82.7
IS-9	SAM3	Hiera-L	71.1	87.9	81.2

ID	Tracker	MOTA	MOTP	IDF1	IDSW	Prec	Recall
TR-5	ByteTrack	84.1	29.8	91.0	590	99.5	85.9
TR-6	StrongSORT	64.2	24.2	78.0	345	99.8	65.2

ID	Model	Pre-train	Acc	F1 _{mac}	Prec	MCC
CL-6	Swin-T	ImageNet	91.7	91.7	92.4	0.873
CL-3	EfficientNetV2-S	ImageNet	91.4	91.4	91.5	0.869

Key Takeaways

- New Benchmark:** PhenoLeaf-TS provides 17,082 RGB images of 318 plants across 21 *Arabidopsis* genotypes for studying individual leaves over time.
- Three Benchmark Tasks:** Establishes baselines for leaf instance segmentation, tracking and growth-stage classification through comparisons of nine segmentation models, six trackers and six classifiers.
- Consistent Leaf Identities:** Colour-coded annotations preserve each leaf's identity across image sequences, supporting the evaluation of segmentation accuracy and tracking consistency during plant growth.

